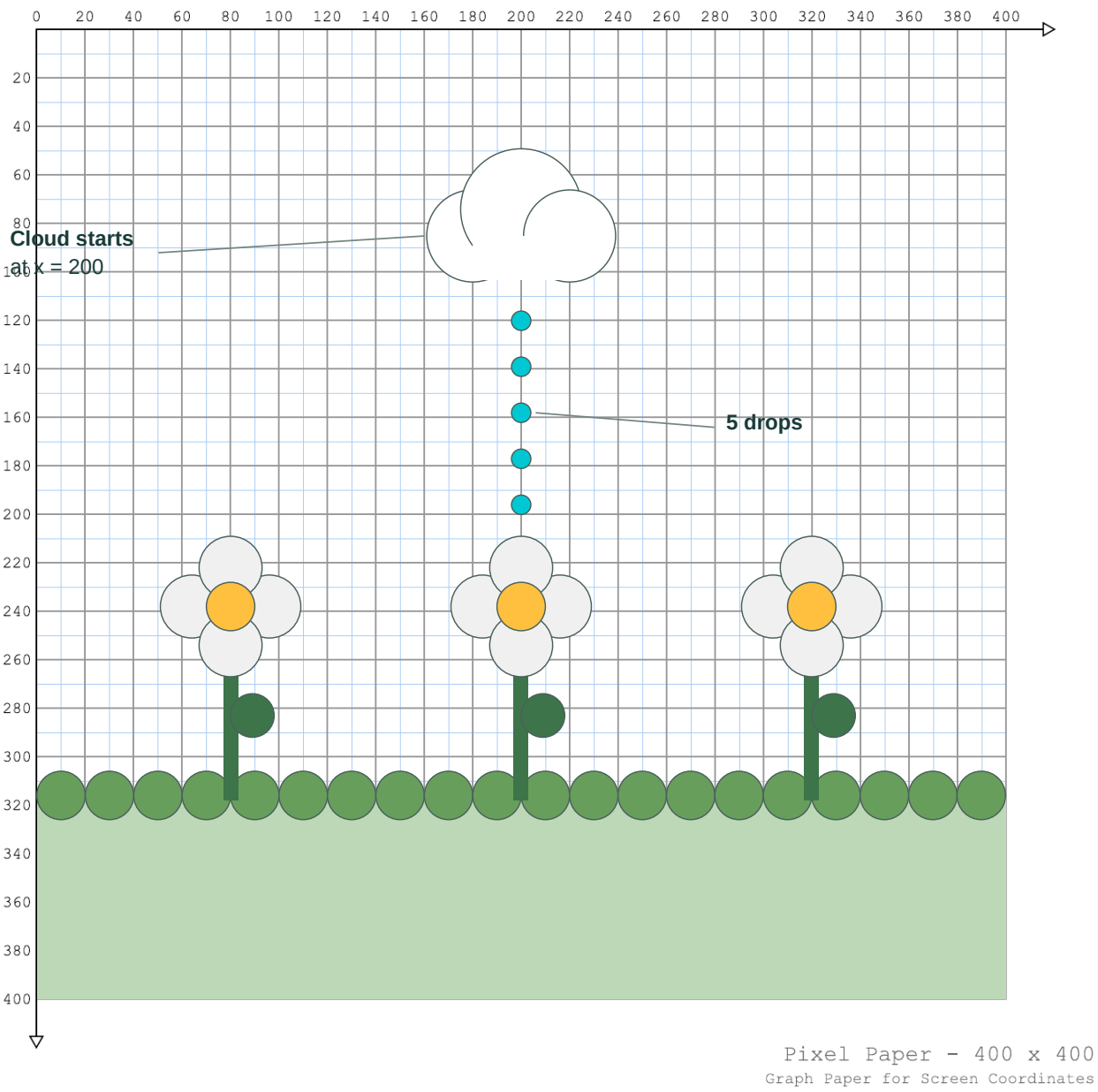
Rain Garden drawing plan

Starting layout on the supplied 400 x 400 pixel paper. Coordinates start at the top left.



Plotting notes

Flower centres: (80, 238), (200, 238), (320, 238). Each yellow centre has radius 10.
Ground begins at y = 316. The three flowers start off-white; watering changes their petals.

Interaction and movement

The cloud shows the selected colour and the horizontal position where rain will land.

01 Move the cloud

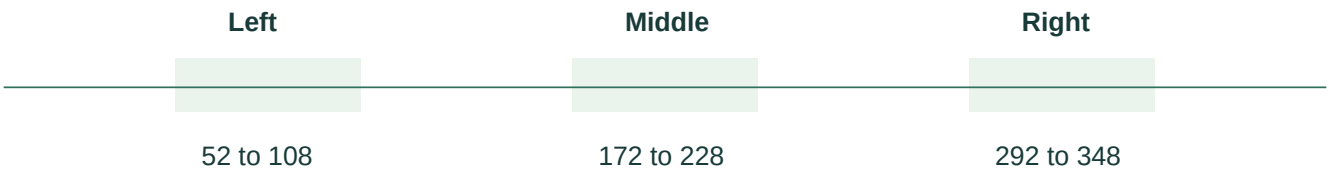
Left and right arrows change x by 160 times Time.DeltaTime each frame.
The cloud centre stops at x = 40 or 360 so its outer circles stay on screen.

02 Choose the rain

Hold W for magenta rain. Release W for cyan. The drops change colour immediately.
A flower changes only when the left mouse button is pressed while the cloud is aligned.

03 Water one flower

Each flower accepts rain from 28 pixels left of its centre to 28 pixels right, inclusive.
The mouse triggers the action; the cloud position chooses which flower receives it.



Example: x = 108 waters the left flower. At x = 109 the click misses all three.

Working out where the cloud must stop

The outer circle centres sit 20 pixels from cloud.x. Each circle has a radius of 19, so the silhouette reaches 39 pixels on either side. Stopping the centre at x = 0 would put its left edge at -39. At x = 40 the left edge is 1; at x = 360 the right edge is 399.

Check the gap between flowers

At x = 140 the cloud is 60 pixels from both neighbours. Each accepts only 28 pixels, so neither should change. The simulated click at x = 140 left all three colours intact.

Programming plan

Three reusable classes share the drawing and interaction work.

Cloud	Stores x and the current rain colour. Move(), ChooseRain(), DrawCloud()
Flower	Stores one centre x and one petal colour per flower. Water(cloud), DrawFlower()
Meadow	Stores groundY. Draws the ground and the row of grass tufts. DrawMeadow()

Repeat a shape with a for loop

Raindrops: y = 120 + drop * 19. For drop = 0 through 4, y is 120, 139, 158, 177, 196.
Grass: x = 10 + tuft * 20. Twenty tufts run from x = 10 through 390, all at y = 316.

Keep each flower independent

Game creates three Flower objects. On a click, each compares cloud.x to its own range.
Only a matching flower copies cloud.rain into petals. The others keep their last colour.

Problem: releasing W could undo a watering

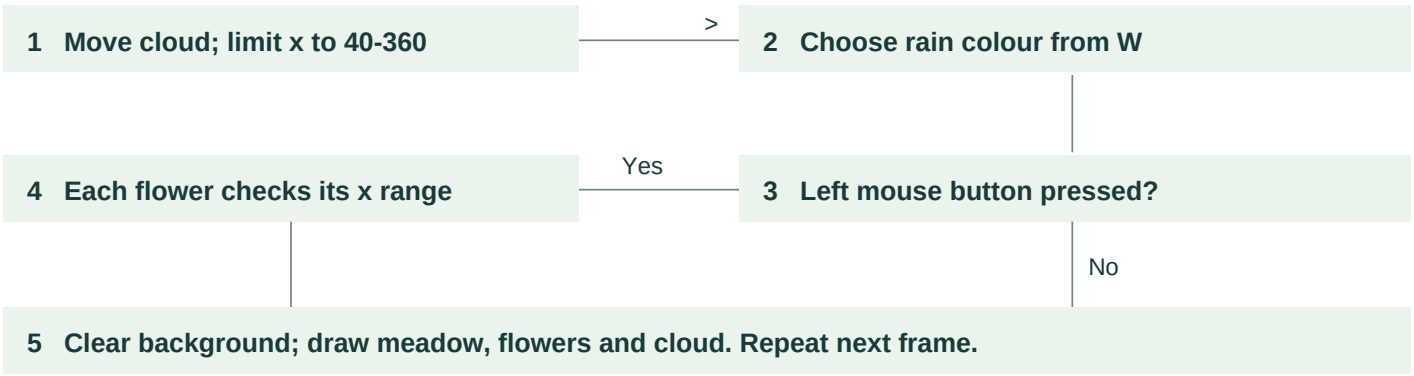
If DrawFlower() read cloud.rain every frame, releasing W would turn the petals cyan.
Instead, Water() copies the colour into that flower only on an aligned click.
In the test, releasing W turned all five drops cyan while the middle flower stayed magenta.

Trace the final pass of the grass loop

At tuft = 19, x = 10 + 19 x 20 = 390. A radius of 10 reaches the right boundary at 400.
At tuft = 20 the condition fails. An extra pass would centre a tuft at 410, off the canvas.

Frame flow and test checklist

Automated check of Game.Update() with simulated inputs and recorded drawing calls.



Test checklist

All six cases passed. Flower colours are reported in left, middle, right order.

☑ Hold each arrow to the edge. The whole cloud should stay visible.

PASS At the two stops, the outer cloud edges reached x = 1 and x = 399 on the 400 px canvas.

☑ Release W and click above the left flower. Only its petals should turn cyan.

PASS Click at x = 80: cyan / off-white / off-white.

☑ Hold W and click above the middle flower. Left stays cyan; middle turns magenta.

PASS W + click at x = 200: cyan / magenta / off-white.

☑ Release W without clicking. Drops should turn cyan; middle petals should stay magenta.

PASS All five drops changed to cyan. Petals stayed cyan / magenta / off-white.

☑ Click halfway between flowers. All three should keep their previous colour.

PASS Click at x = 140: cyan / magenta / off-white. No flower changed.

☑ Return to the middle and click without W. The same flower should change back to cyan.

PASS Click at x = 200: cyan / cyan / off-white. Only the middle flower changed.